Stage 2 Report

Claire Standley

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Topic and Motivation:

Direct-to-consumer (DTC) agricultural markets, such as farmers markets, on farm stands, and other forms of local food marketing, have grown substantially in the United States over the past decade. These markets allow farmers to capture a larger share of the consumer food dollar and may provide important opportunities for small and beginning farmers who face barriers to entry in traditional agricultural supply chains. Many papers have found that first time farmers and younger farmers are more likely to participate in value added agricultural often time as a means to provide extra income within the first few years to be able to keep the farm operational.

One factor that may influence the growth of local food markets is the legal environment surrounding agritourism. Agritourism includes activities such as farm tours, pumpkin patches, corn mazes, and on farm events that may attract visitors to agricultural operations. Other papers have observed clustering effects regarding agritourism and direct to consumer sales in the United States and found these two methods of income to be closely related in most cases. However, farms that host visitors may face liability risks if guests are injured while visiting the property.

Starting with Kentucky in 2002, many states adopted agritourism liability laws which limit the legal liability of farmers hosting agritourism activities under certain conditions. By reducing the risk of lawsuits, these laws may encourage farmers to expand agritourism activities and attract more visitors. Increased visitor traffic also increases opportunities for farms to sell products directly to consumers. This project will attempt to examine whether the adoption of agritourism liability laws is associated with changes in the development of local food marketing channels, particularly direct to consumer agricultural markets. Understanding the relationship between agritourism policy and local food systems is important for policymakers interested in supporting rural economic development, agricultural diversification, and farm profitability. Additionally, a (better) study of such could provide evidence of an unintended benefit of expansion of limited liability laws to the agricultural community and specifically an opportunity to help with endeavors that can lower barriers to entry for young agriculturists.

Datasets:

USDA Farmers Market Directory (USDA Local Food Directories) – provided by the Agricultural Marketing Service (AMS) and provides a nationwide listing of farmers markets across the United States. The dataset includes information about individual markets, their location, and some characteristics of their operations. Variables include market name, city and state, zip code, lat and long, market website, and some vendor information. The timespan in from 2020 to early 2026 being updated periodically. Markets are added over time as they register with the USDA. The datasets include farmers markets across the entire United States, with geographic information that allows markets to be mapped to counties. Each state has several entries as a representative sample. This dataset will allow me to construct a county-by-year measure of the number of farmers markets. Farmers markets serve as a proxy for the development of local marketing channels and direct-to-consumer sales opportunities.

Agritourism Liability Law Adoption by State (Agritourism – National Agricultural Law Center) – constructed from publicly available legal sources documenting state agritourism liability statues. Many states have adopted laws limiting the liability of farms that host agritourism activities starting as far back as 2004. The website I found provides links to the listing of all current statues for each state and through publicly available records first adoption date for such laws as well as any amendment dates can be obtained. The variables include state, year agritourism liability law was first adopted, any amendment years to the law, indicator variable for whether a state has an agritourism liability law in a given year. (I will kind of have to create this based on information on this site searching each statue to obtain adoption and amendment years – willing to bear that cost should take me too long, I plan to get this finished over spring break, plus will be technically publicly available through my public GitHub repository afterwards). The timespan is from 2004 through current with some state still not having established laws on agritourism liability. The dataset includes all 50 United States, allowing comparisons between states that adopted laws at different times. This dataset will provide the policy variables used in the analysis. The adoption year will be used to create a treatment indicator that identifies when a state begins operating under agritourism liability protection.

Census of Agriculture (USDA - National Agricultural Statistics Service - Census of Agriculture) – Direct-to-consumer Sales – The USDA Census of Agriculture is a comprehensive survey of U.S. farms conducted every 5 years. The census collects detailed information about farm operations, production practices, farm finances, and marketing channels. One category report in the census is DTC agricultural sales income which measures the value of farm products sold directly to consumers without intermediaries. It also includes income for agritourism at the county level as well which can also be used to explore heterogeneity in the sample. Variables include state, county, value of direct-to-consumer sales/income, agritourism income, total number of farms in a given county, farm operator demographics. The relevant time span will be 2002-2022 (the most recent year) in 5-year intervals. It is a United States national dataset and covers every state at the county level. County-level data allows the analysis to examine local variation to DTC agricultural activity. Due to this being census data and only being done every 5 years I don't plan to use this data as my primary data source as this would be a miss match with being able to accurately record pre and post policy. However, if the data listed in data set 1 is not sufficient, I have a backup dataset that will not allow for the most accurate identification of variation but is still something. The datasets will be combined using geographic identifiers (state and county). Farmers markets, on farm markets, and the agritourism from the USDA directory can be assigned to counties and states using their location information. These county or state observations can then be matched with the agritourism liability law dataset and the census data using the state identifiers. This will allow the project to examine whether the number of farmers markets changes in states after agritourism liability laws are adopted.

Data Processing:

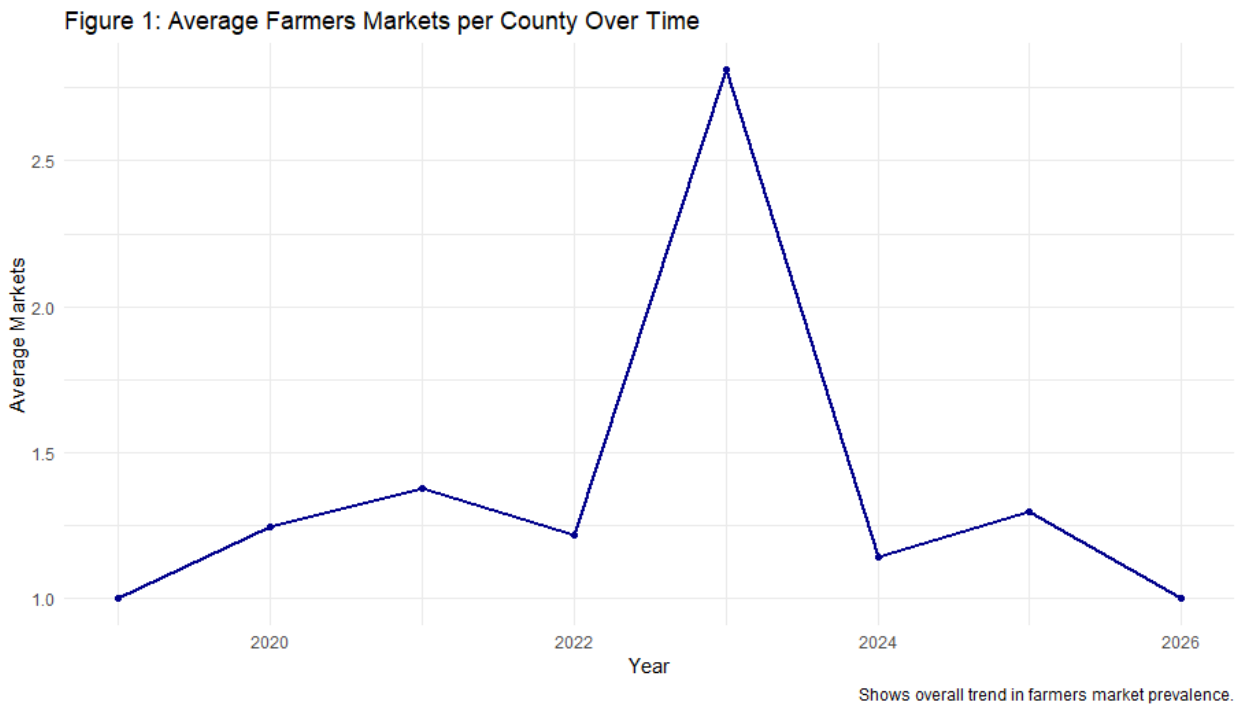
I started by loading in my 3 data sets listed above (USDA census data, ag law data, and farmers market data). All data sets were cleaned per the data cleaning checklist from the course. They were cleaned for consistency in variable naming and formatting first. I then standardized capitalization, and converted years to numeric. Duplicate rows and empty observations were removed to avoid double counts. I merged the data sets using county, state, and year as the primary keys for county-level data, while the state laws was expanded into a panel for each state-year combination. This allows me to capture variation in farmers market prevalence and farm participation over time while linking outcomes to the timing of the state agritourism laws. The unit of analysis for the merged data sets is the county-year for the farmers market and census data, with state-level law characteristics attached to each county-year observation.

Transformations, Missing, and Extreme Values:

Several transformations were applied before moving into visualization. First due to the long tails presented by the histogram I log transformed number of markets and number of farms with log1p (because some

values were 0). This reduced skewness and stabilized my variance. Extreme values therefore are not overly influential in my data but don't get entirely dropped either. For categorical grouping, the number of statutory codes in state law was binned into meaningful groups at natural breaks to see how the law complexity could impact direct-market operations. Missing data was address in cases of the census data such as (D) which were converted to NA values as those values were not listed therefore not applicable as a value. The D is suppressed data withheld for individual reasons therefore could not be used. For the variable capturing farms with value-added sales, remaining missing values were replaced with 0 where appropriate (no instances that I could see but coded in case of values). I checked extreme values at the 50th, 90th, 95th, and 99th percentile to ensure no removal was necessary. Since our log transformation had already been done and no 99th percentile values were substantially large I did not remove values for this reason. Hopefully every decision I made was convey well in my reasoning behind the decision and that the data set cleaned reflects true participation in value added agriculture/programs.

Visualizations:



This figure illustrates the trend in average farmers markets in counties over time. It shows a clear increase in farmers markets in 2023 on average. This could be due to a push towards registration with the given site as individuals are the ones that provide the data. After registering in 2023 individuals did not have to register again so likely we saw a push for registration in 2023 and then the apparent fall is due to those farmers markets still being open but not necessarily re-registering.

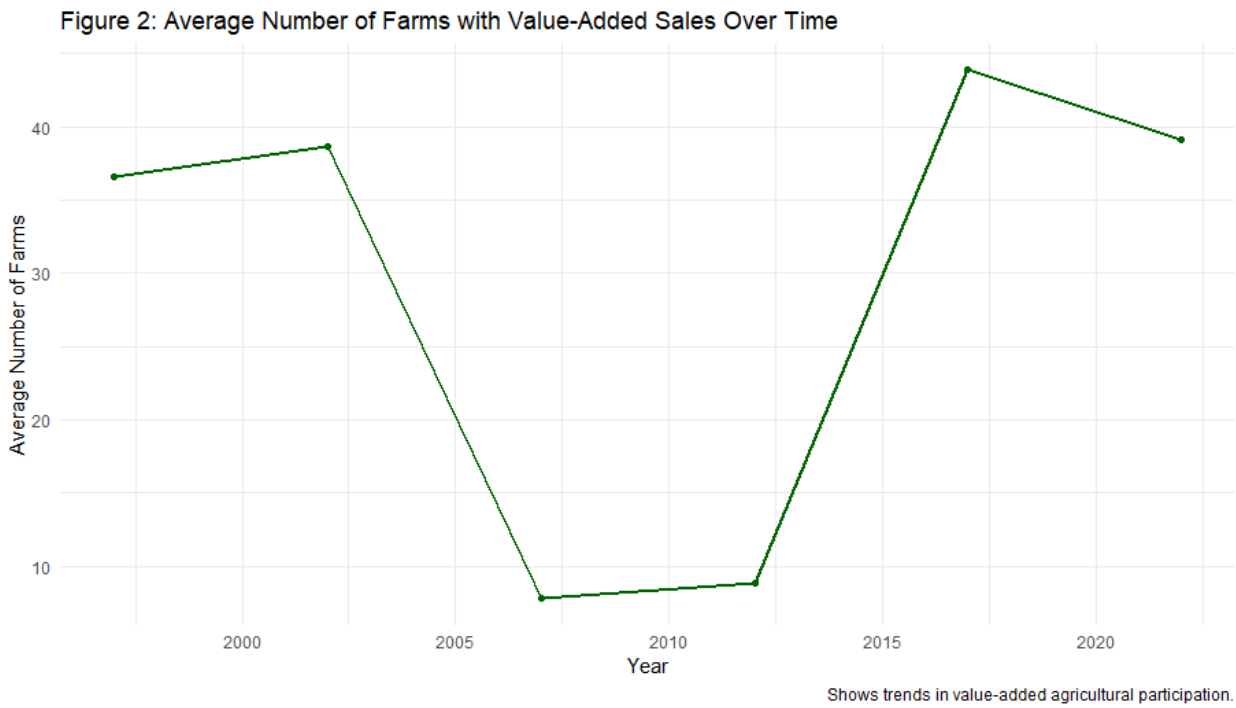


Figure 2 uses the USDA census data to measure average number of farms with value-added sales over time. It shows a large decrease in overall number between 2007 and 2012. This is likely due to the fact that definition of that variable was change in these years leading to fewer farms actually fitting into that category. We see an increase again in 2017 when the variable was once again redefined to be more inclusive. Overall we do see a larger number today in comparison to the early 2000 showing some potential growth although it is hard to tell the overall trend due to 2007 and 2012.

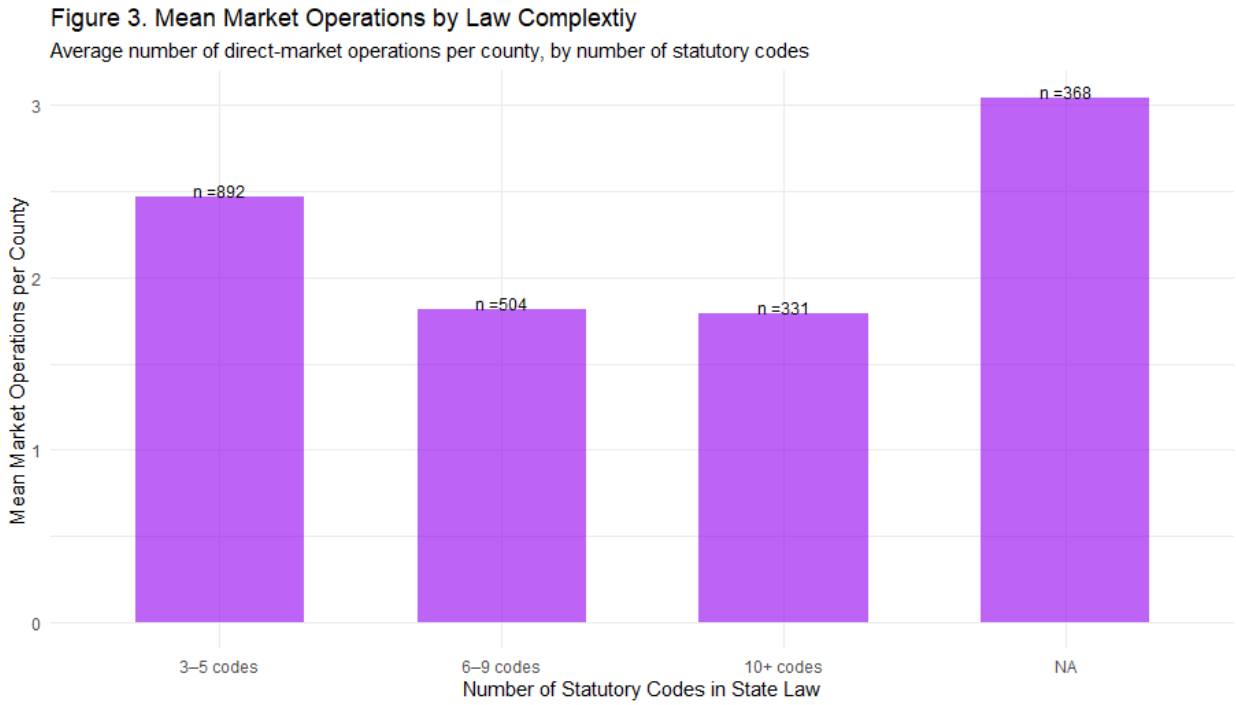


Figure 3 shows the relationship between code complexity and mean market operation per county. There are more data entries for those state who never adopted as the NA values. Most states have 305 codes as

evidenced by 892 entries. The mean market operations decreases slightly by more complex codes (those that have more codes) but by only about 1.

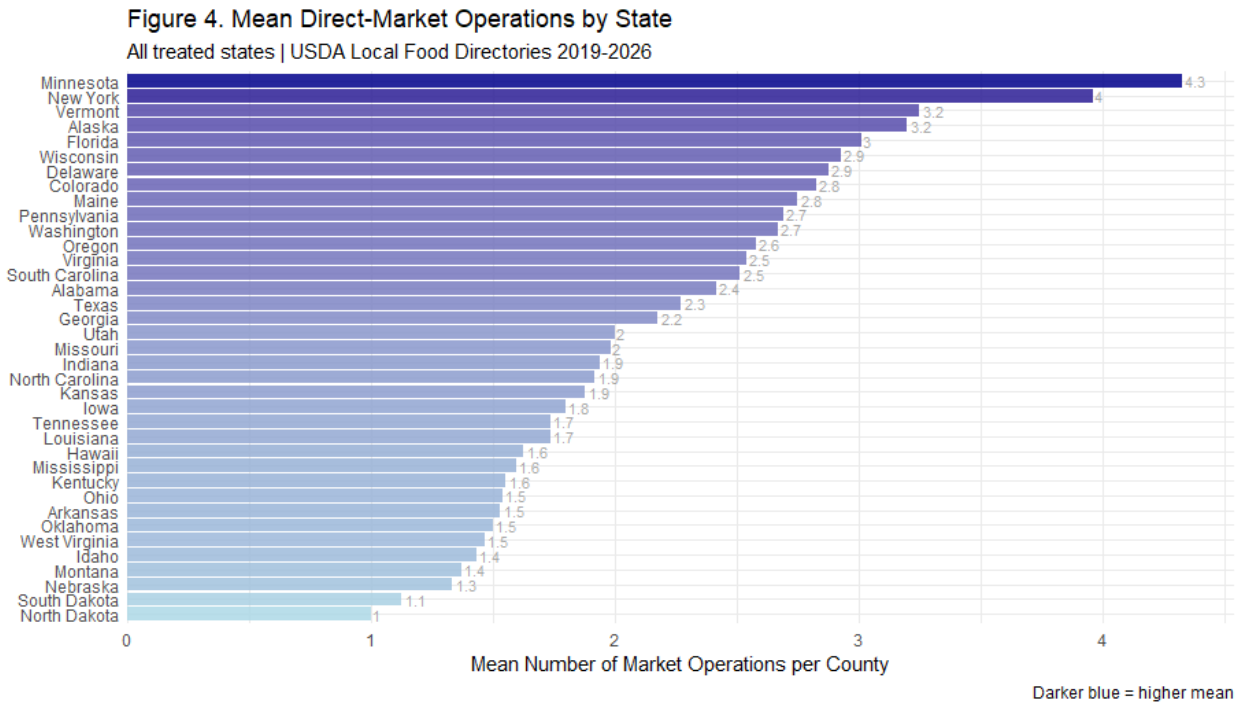
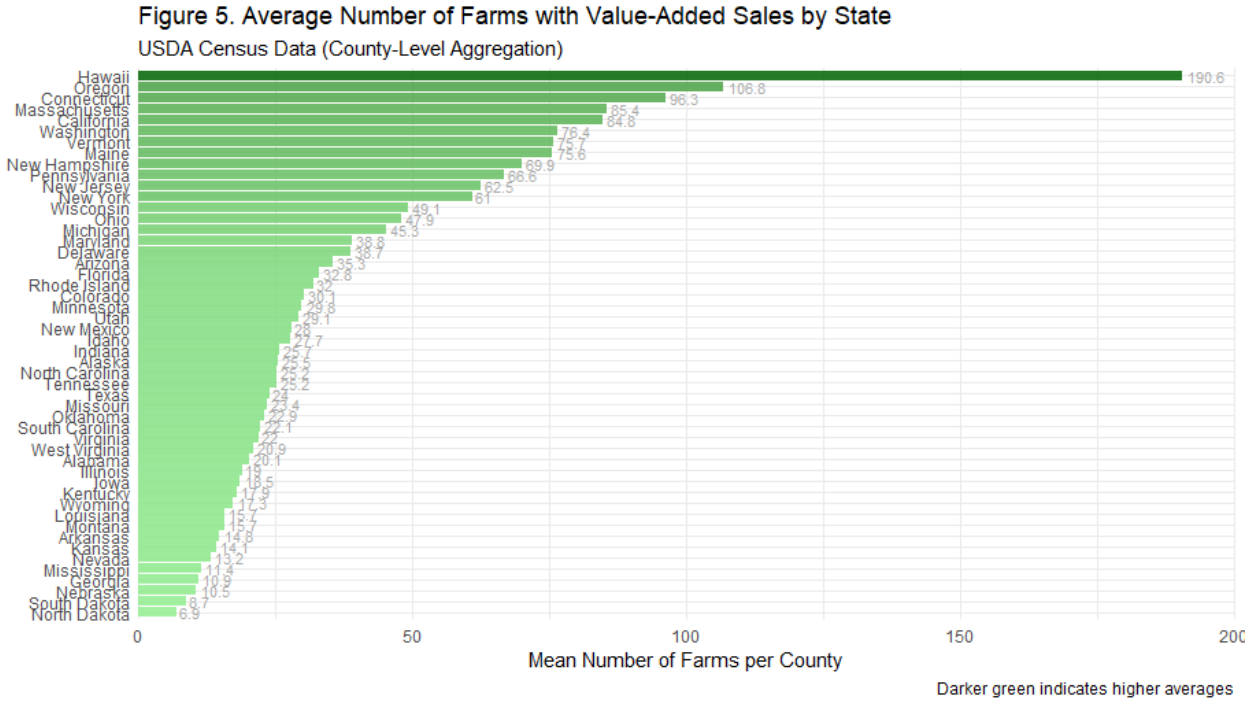


Figure 4 shows by state the mean number of operations of farmers markets. Minnesota is the state with the most during the time period provided having on average 4.3 farmers markets per county. The state with the least is North Dakota with only around 1 farmers market.



This graph shows using the census data instead the average number of farms by state. Using the census data Hawaii is now the largest with Minnesota falling in the middle. This leads me to believe that many

individuals in Hawaii just may not know about the program or register their market. Alternatively it could be that farmers market is not a great proxy for value added and that the Hawaii direct to consumer is not in the form of farmers market. However our states towards the bottom North Dakota and South Dakota did not change suggesting that those states truly do not participate in value added as much.